

# Instrumental Eigenmode Spectroscopy of a Holographic Einstein–Dilaton Bulk Using Optical Clocks

An ex-ante, no-feedback protocol with null-shuffle controls and symmetric local vs network readout

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## Abstract

An instrument-level protocol is specified to test whether optical-clock time series exhibit an auditable response to bulk eigenmode perturbations of a frozen Einstein–Dilaton background under an ex-ante, no-feedback sweep. An integrated temporal dictionary,  $t_{\text{int}}(\epsilon; n) = \tau_R \int_{z_{\text{src}}}^{z_{\text{dst}}} \exp[-(A(z) + \epsilon \Phi_n(z))] dz$ , is recomputed over a fixed list of perturbation amplitudes  $\epsilon$  and eigenmodes  $\Phi_n(z)$ . For each  $(n, \epsilon)$ , a sliding-window analysis reports SNR and demodulated phase coherence at a pre-registered target frequency. At  $\epsilon = 0$  (control), all mode labels are identical by construction (same dictionary). At  $\epsilon = 0.02$  (first nonzero perturbation), the dictionary becomes mode-dependent and yields mode-dependent phase-coherence structure under the same fixed decision rule.

## 1 Scope and non-claims

- The objective is an *instrumental* test: whether a fixed, auditable operator pipeline produces stable SNR and phase-coherence metrics under an ex-ante protocol.
- No claim of feedback control, post-hoc tuning, parameter fitting, or maximization is made; the sweep is pre-registered and executed as written.
- The reported observables are *instrument-level spectroscopy* diagnostics; no claim of a new interaction, communication channel, or causal propagation is required.

## 2 Null-space write modes: from instrumental operator to bulk eigenmodes

Let  $\mathcal{O}$  denote an (abstract) instrument response operator associated with the fixed analysis pipeline and decision rule (mapping a bulk/state perturbation into a readout statistic under a fixed protocol). An SVD separates observable directions from directions that are invisible to the readout:

$$\mathcal{O} = U \Sigma V^\top, \quad \Sigma = \text{diag}(\sigma_1, \dots, \sigma_r, 0, \dots, 0). \quad (1)$$

The null-space is the set of perturbations that do not excite the direct readout channel:

$$\mathcal{N} = \ker(\mathcal{O}) = \text{span}\{v_{r+1}, \dots, v_n\}. \quad (2)$$

To avoid an arbitrary choice inside  $\mathcal{N}$ , a physically selected direction can be defined by a minimal-energy criterion,

$$v_j = \arg \min_{v \in \mathcal{N}} \mathcal{E}(v), \quad (3)$$

and mapped to a bulk profile by projecting onto background parameter directions,

$$\psi_j(z) = \sum_i v_{j,i} \frac{\partial \Phi(z)}{\partial \alpha_i}. \quad (4)$$

The bulk profile is then constrained by the physical eigenvalue problem (Sturm–Liouville form),

$$(-\square + V''(\Phi_0)) \psi_j = \omega_j^2 \psi_j, \quad (5)$$

which defines a triad

$$\boxed{\text{write mode } j \equiv (v_j, \psi_j, \omega_j)}. \quad (6)$$

The write-mode hypothesis is that geometry can change without inducing a local force in the readout channel:

$$\boxed{\delta a^\mu = 0, \quad \delta g_{\mu\nu} \neq 0}. \quad (7)$$

This provides the operator-level motivation for using bulk eigenmodes as physically selected write-mode profiles rather than an arbitrary basis. In this work, the injected profiles  $w(z)$  are taken directly as the frozen bulk eigenmodes  $\Phi_n(z)$ .

### 3 Operator and dictionary definition

#### 3.1 Bulk eigenmodes

Let  $\Phi_n(z)$  denote the first ten eigenmodes of a self-adjoint bulk Sturm–Liouville problem defined on the frozen background. Each  $\Phi_n$  is treated as a fixed perturbation profile over the bulk coordinate  $z$ .

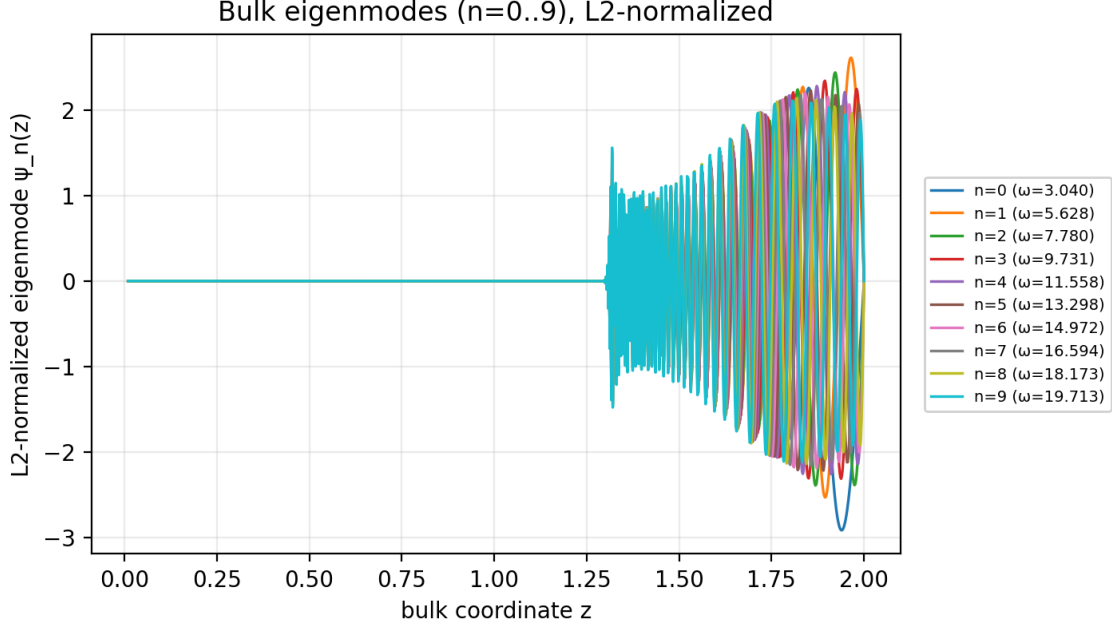


Figure 1: Bulk eigenmode profiles used as perturbation weights. Curves are  $L^2$ -normalized under  $\langle \psi_n, \psi_n \rangle = \int \psi_n(z)^2 dz = 1$  (numerical quadrature on the frozen grid). The legend reports  $\omega_n$  for  $n = 0..9$  in the spectrum units associated with the frozen operator.

### 3.2 Integrated temporal dictionary

The only time observable used for the response sweep is the accumulated (phase) time along the bulk coordinate:

$$t_{\text{int}}(\epsilon; n) = \tau_R \int_{z_{\text{src}}}^{z_{\text{dst}}} \exp[-(A(z) + \epsilon \Phi_n(z))] dz, \quad (8)$$

with  $\tau_R$ ,  $(z_{\text{src}}, z_{\text{dst}})$ , and numerical quadrature grid fixed for all runs. Mode dependence enters *only* through  $\Phi_n(z)$  and  $\epsilon$ .

## 4 Ex-ante protocol (no feedback)

**Fixed sweep (control and spectroscopy points).** A finite list  $\epsilon \in \{-0.05, -0.02, 0, 0.02, 0.05\}$  is fixed before analysis. No adaptive updates are permitted based on intermediate results. The point  $\epsilon = 0$  is an explicit *control*: the dictionary is identical for all mode labels. The first nonzero point  $\epsilon = 0.02$  is the minimal perturbation at which *mode spectroscopy* becomes well-defined.

**Definition (ex ante).** “Ex ante” means that  $\epsilon$ , the SNR threshold, and the windowing scheme are fixed *before* inspecting results (no post-hoc adjustment); it is equivalent to “pre-registered”.

**Windowing.** For each dataset and each  $\epsilon$ , a sliding window is applied with fixed window and step durations (set separately for local and network groups). For each window, an FFT bin at a target frequency is evaluated, recording: (i) SNR at the target bin and (ii) phase at the target bin. To enable circular statistics across windows, a demodulated phase is used:

$$\phi_{\text{demod}} = \text{Arg}[\exp(i(\phi - 2\pi f_{\text{target}} t_{\text{start}}))]. \quad (9)$$

**Decision rule.** A pre-registered SNR threshold of 10 dB defines “passing windows”. Phase coherence metrics are computed *only* on passing windows; windows are not otherwise excluded.

**Null shuffles.** For each window, null-shuffle controls are generated by shuffling within-window samples a fixed number of times. Null means and standard deviations are recorded and displayed on the same axes as the data.

## 5 Metrics

For each dataset:

- max SNR: maximum SNR across windows (at the target bin),
- $f_{\text{peak}}$ : window-local peak frequency (for diagnostic stability),
- $\phi_{\text{demod}}$ : demodulated phase on passing windows,
- $(R, \sigma_\phi)$ : circular phase coherence and circular dispersion on passing windows.

For each group and each  $(n, \epsilon)$ , dataset-level metrics are aggregated deterministically into:

- pass rate  $p$ : fraction of datasets with max SNR  $\geq$  threshold,
- max SNR<sub>max</sub>: maximum of dataset-level max SNR,
- $R_{\text{med}}$ : median phase-coherence  $R$  over datasets with at least one passing window.

## 6 Results

### 6.1 Control vs mode-dependent dictionary

At  $\epsilon = 0$ , Eq. (8) is mode-independent; all mode labels correspond to the same dictionary. At  $\epsilon = 0.02$ ,  $t_{\text{int}}$  varies across modes by  $\mathcal{O}(10^{-3})$ – $\mathcal{O}(10^{-2})$  relative, confirming that the sweep is not a mere relabeling (Table 3).

### 6.2 Group-level response heatmaps

Table 1 and Figures 2–3 summarize mode response at fixed  $\epsilon$  for the two readout groups. Heatmaps report aggregated quantities without exclusion of datasets.

Table 1: Pass-rate summary at the fixed SNR threshold. For the present evaluation set,  $p$  is mode-independent under the fixed decision rule; per-mode values are listed in Table 2.

| Group       | $p(\epsilon = 0)$ | $p(\epsilon = 0.02)$ |
|-------------|-------------------|----------------------|
| Local (L)   | 1.00              | 1.00                 |
| Network (N) | 0.00              | 0.00                 |

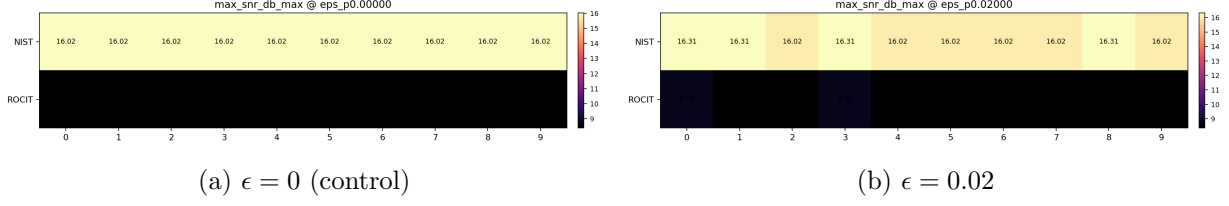


Figure 2: Group-level max SNR<sub>max</sub> (maximum dataset max SNR) at fixed  $\epsilon$ .

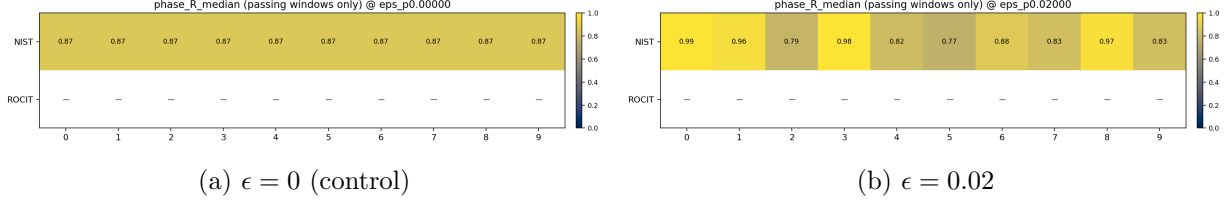


Figure 3: Median phase-coherence  $R_{\text{med}}$  computed across datasets that contain at least one passing window. Cells are blank when no passing windows exist.

### 6.3 Aggregated mode summary at $\epsilon = 0.02$

Table 2 lists a compact mode $\times$ group $\times\epsilon$  summary for  $\epsilon = 0$  (control) and  $\epsilon = 0.02$  (first spectroscopy point).

Table 2: Compact mode response summary at  $\epsilon \in \{0, 0.02\}$  for the local (L) and network (N) groups. Here  $p$  is pass rate, max SNR is in dB, and  $R$  is phase coherence on passing windows.

| Mode | $\epsilon$ | $p_L$ | max SNR <sub>L</sub> | $R_L$ | $p_N$ | max SNR <sub>N</sub> | $R_N$ |
|------|------------|-------|----------------------|-------|-------|----------------------|-------|
| 0    | 0          | 1.00  | 16.02                | 0.873 | 0.00  | 8.37                 | —     |
| 1    | 0          | 1.00  | 16.02                | 0.873 | 0.00  | 8.37                 | —     |
| 2    | 0          | 1.00  | 16.02                | 0.873 | 0.00  | 8.37                 | —     |
| 3    | 0          | 1.00  | 16.02                | 0.873 | 0.00  | 8.37                 | —     |
| 4    | 0          | 1.00  | 16.02                | 0.873 | 0.00  | 8.37                 | —     |
| 5    | 0          | 1.00  | 16.02                | 0.873 | 0.00  | 8.37                 | —     |
| 6    | 0          | 1.00  | 16.02                | 0.873 | 0.00  | 8.37                 | —     |
| 7    | 0          | 1.00  | 16.02                | 0.873 | 0.00  | 8.37                 | —     |
| 8    | 0          | 1.00  | 16.02                | 0.873 | 0.00  | 8.37                 | —     |
| 9    | 0          | 1.00  | 16.02                | 0.873 | 0.00  | 8.37                 | —     |
| 0    | 0.02       | 1.00  | 16.31                | 0.985 | 0.00  | 8.74                 | —     |
| 1    | 0.02       | 1.00  | 16.31                | 0.958 | 0.00  | 8.37                 | —     |
| 2    | 0.02       | 1.00  | 16.02                | 0.793 | 0.00  | 8.35                 | —     |
| 3    | 0.02       | 1.00  | 16.31                | 0.985 | 0.00  | 8.74                 | —     |
| 4    | 0.02       | 1.00  | 16.02                | 0.822 | 0.00  | 8.35                 | —     |
| 5    | 0.02       | 1.00  | 16.02                | 0.767 | 0.00  | 8.35                 | —     |
| 6    | 0.02       | 1.00  | 16.02                | 0.877 | 0.00  | 8.37                 | —     |
| 7    | 0.02       | 1.00  | 16.02                | 0.831 | 0.00  | 8.35                 | —     |
| 8    | 0.02       | 1.00  | 16.31                | 0.970 | 0.00  | 8.37                 | —     |
| 9    | 0.02       | 1.00  | 16.02                | 0.830 | 0.00  | 8.35                 | —     |

## 7 Limitations and falsification criteria

- Mode-dependent response in the local group is conditional on the adopted operator pipeline and its fixed target-frequency rule.
- A positive claim of portability across groups would require threshold-crossing and phase stability in independent network datasets under the same fixed rules; failure to meet this is a falsification of portability under the tested protocol.
- The reported observables do not by themselves identify a microscopic coupling mechanism; they are operator-level response diagnostics.

Table 3: Integrated dictionary sensitivity at  $\epsilon = 0.02$ :  $t_{\text{int}}(0.02)/t_{\text{int}}(0)$  per mode, demonstrating that the dictionary is mode-dependent at the first nonzero perturbation.

| Mode | $t_{\text{int}}(0)$ [s] | $t_{\text{int}}(0.02)$ [s] | $t_{\text{int}}(0.02)/t_{\text{int}}(0)$ |
|------|-------------------------|----------------------------|------------------------------------------|
| 0    | 1000.84                 | 1004.86                    | 1.004013                                 |
| 1    | 1000.84                 | 1002.91                    | 1.002067                                 |
| 2    | 1000.84                 | 999.67                     | 0.998831                                 |
| 3    | 1000.84                 | 1004.99                    | 1.004146                                 |
| 4    | 1000.84                 | 1000.07                    | 0.999228                                 |
| 5    | 1000.84                 | 999.34                     | 0.998496                                 |
| 6    | 1000.84                 | 1000.89                    | 1.000053                                 |
| 7    | 1000.84                 | 1000.20                    | 0.999360                                 |
| 8    | 1000.84                 | 1003.34                    | 1.002501                                 |
| 9    | 1000.84                 | 1000.19                    | 0.999347                                 |

## Data & Reproducibility

All figures and tables in this work are generated from a minimal set of machine-readable JSON files: (i) the aggregated mode  $\times$  group  $\times$   $\epsilon$  table (`mode_response_rows.json`), (ii) representative window-level reports (`window_sweep_report.json`) for two modes and two readout groups, and (iii) the integrated temporal dictionary (`temporal_dictionary_integrated.json`). These files are sufficient to reproduce all numerical values and figures in the manuscript under the fixed ex-ante protocol.

Machine-readable verification artifacts supporting the quoted numerical values will be uploaded to [https://github.com/RAPIDENN/HOLO\\_runner](https://github.com/RAPIDENN/HOLO_runner).